செம்மல் | Semmal

4-minute video: plan, shot list and voiceover

Goal: open on the problem (first 30 seconds), then show only the deliverables that matter, each proven on screen. ~560 words of voiceover at about 140 words per minute.

0:00–0:28 is the problem. 0:28–0:40 names what we built. Then each of these is proven on screen:

- **Robots are their own agents**: they price the job, hire AI agents and pay them in 0G.
- **0G is the brain at every checkpoint**: private TEE inference on 0G Private Computer, request ID per decision.
- **The robot signs its own record**: episode bundle on 0G Storage, indexed on Arkiv, owned by the robot's wallet.
- **Safe by design**: no answer from 0G means a safe stop, never a guess; motor faults stop it with no network.
- **Scores choose who gets work**: robots and agents ranked by scores anyone can recompute from Arkiv.

What you need before recording

Item	Detail
Robot PC (Ubuntu)	`scripts/robot/scenario.sh house` running (Gazebo window visible), `pnpm start house` linked to the platform; toys and spill reset (they reset at each job start)
Platform (Mac)	`pnpm dev:api` + web on **http://localhost:3000** (avoids MetaMask's vercel.app warning); or a custom domain
Wallets	Two browser profiles: **customer** and **robot owner**, both signed in. Customer: Rabdi tGLM (booking write) + Galileo 0G (to pay). Robot wallet: 0G to pay agents, tGLM for Arkiv writes
Agent	A vision agent published in **private** mode on `0gm-1.0-35b-a3b`; 0G Router deposit funded
Fault demo	`scripts/robot/inject_fault.sh` ready in a terminal (left motor overcurrent)
Recording	1920×1080, browser zoom 110%, bookmarks hidden, notifications off, mic test. Record each segment separately; cut long waits (a 0G call takes about 20–25 s; travel uses the fast-forward button)
Captions / annotations	Add the on-screen labels in the "Caption" column below in your editor (big, 2–3 s each). Blur nothing: there are no secrets on screen

Shot list and voiceover

Time	On screen	Caption (annotation)	Voiceover
0:00–0:28	Black screen, then the home page hero fades in under the voice	Robots work for hire. Who checks them?	"Robots are starting to work for hire: cleaning homes, inspecting warehouses. They act on what an AI tells them. But nobody can see what that AI was shown, whether it stayed private, who paid for it, or whether the decision was real. When the AI is down, the robot guesses. And its track record belongs to whoever runs the platform."
0:28–0:40	Fast montage (2 s each): How it decided (request ID) → What went to Arkiv (task_score 100) → SAFE_STOP	Private 0G decisions · signed record · safe stop	"Semmal fixes that. The robot decides with a private model on 0G, signs its own record on 0G Storage and Arkiv, and stops safely instead of guessing."
0:40–1:05	/quote: pick a site on the map, robots ranked; open the robot's score; Book (wallet approval card → wallet)	Ranked by score · booking signed by your wallet, on Arkiv	"A customer picks a spot on the map. Robots that cover it are ranked by their score. Booking writes an entity on Arkiv, signed by the customer's own wallet. There is no platform key anywhere."
1:05–1:25	Robot's offer appears; owner profile: requests → Approve	The robot prices it itself	"The robot works out its own price from live agent prices on Arkiv and 0G model costs, signs the quote and publishes it. Its owner approves the job."
1:25–1:45	Map: robot travelling (fast-forward) → 'Arrived: confirm' → Pay 0G	Paid in native 0G on 0G Galileo	"The robot travels to the site. The customer confirms it has arrived and pays the quote in native 0G. The robot checks the payment on chain before it starts."
1:45–2:25	Gazebo: robot drives to the living room; cut to How it decided: hired agent, payment tx, model, private mode, request ID, answer 'toys on floor', checks ✓; back to Gazebo: toys move to the bin	Hires and pays the agent · private TEE inference · request ID · it clears the space itself	"At each room it takes one camera frame and hires a vision agent, paying the agent's owner in 0G. The agent runs a private model on 0G Private Computer. Here's the request ID, the model, private mode and

			the answer: toys on the floor, 95 percent. The robot checks the format, the action and the confidence, then does the job itself: it picks the toys up and puts them in the bin."
2:25–2:40	Booking page: Extra stops → choose bedroom → 'added'	Ask for more during the job: the robot decides	"While it works, the customer can ask for another room. The robot decides: only if the place fits the job and what was paid still covers the extra check."
2:40–3:05	Terminal: inject_fault.sh → Gazebo robot stops instantly; dashboard: E_STOP, incident signed and queued	Safety on the robot: no network, no AI needed	"Safety never waits for the cloud. Here a motor overcurrent is detected and the robot stops immediately, with no network and no AI. The incident is signed and queued, then written to 0G Storage and Arkiv. And if 0G stops answering, the robot holds, retries once, then stops safely. Nothing is made up."
3:05–3:35	Orders → What went to Arkiv: entity link, indexed columns, task_score 100, score table, agent scores, robot score maths; explorer page	Read back from Arkiv · score: 100 · anyone can recompute	"When the job ends, the robot scores itself: stops done, valid 0G decisions, findings handled, on time. That score and the agents' scores are in its signed bundle on 0G Storage and its entity on Arkiv. This screen reads it straight back from Arkiv, and anyone can recompute it with one query."
3:35–3:50	Scores table: score + selection for robots and agents	Scores pick who gets hired	"Those scores rank robots for customers and agents for robots. Bad ones drop out, and a safe stop never counts against a robot."
3:50–4:00	Logo end card: செம்மல் · Semmal · github.com/jonathanvineet/semmal · Telegram/X jonathan_vineet	Semmal: built at ETHGlobal Mumbai	"Semmal: Tamil for a person of eminence. Robots that earn that name job by job. Built on 0G and Arkiv."

Full voiceover transcript

[0:00] Robots are starting to work for hire: cleaning homes, inspecting warehouses. They act on what an AI tells them. But nobody can see what that AI was shown, whether it stayed private, who paid for it, or whether the decision was real. When the AI is down, the robot guesses. And its track record belongs to whoever runs the platform.

[0:28] Semmal fixes that. The robot decides with a private model on 0G, signs its own record on 0G Storage and Arkiv, and stops safely instead of guessing.

[0:40] A customer picks a spot on the map. Robots that cover it are ranked by their score. Booking writes an entity on Arkiv, signed by the customer's own wallet. There is no platform key anywhere.

[1:05] The robot works out its own price from live agent prices on Arkiv and 0G model costs, signs the quote and publishes it. Its owner approves the job.

[1:25] The robot travels to the site. The customer confirms it has arrived and pays the quote in native 0G. The robot checks the payment on chain before it starts.

[1:45] At each room it takes one camera frame and hires a vision agent, paying the agent's owner in 0G. The agent runs a private model on 0G Private Computer. Here's the request ID, the model, private mode and the answer: toys on the floor, 95 percent. The robot checks the format, the action and the confidence, then does the job itself: it picks the toys up and puts them in the bin.

[2:25] While it works, the customer can ask for another room. The robot decides: only if the place fits the job and what was paid still covers the extra check.

[2:40] Safety never waits for the cloud. Here a motor overcurrent is detected and the robot stops immediately, with no network and no AI. The incident is signed and queued, then written to 0G Storage and Arkiv. And if 0G stops answering, the robot holds, retries once, then stops safely. Nothing is made up.

[3:05] When the job ends, the robot scores itself: stops done, valid 0G decisions, findings handled, on time. That score and the agents' scores are in its signed bundle on 0G Storage and its entity on Arkiv. This screen reads it straight back from Arkiv, and anyone can recompute it with one query.

[3:35] Those scores rank robots for customers and agents for robots. Bad ones drop out, and a safe stop never counts against a robot.

[3:50] Semmal: Tamil for a person of eminence. Robots that earn that name job by job. Built on 0G and Arkiv.

Deliverables this video must show (highlight them)

Track	Must show	At
0G Physical AI	**0G call inside the decision loop**	1:45–2:25
0G Physical AI	**Episode log on 0G Storage with a request ID per decision**	3:05–3:35
0G Physical AI	**Replayable simulation** (the Gazebo world, rerun from scratch)	1:45, 2:40

0G Physical AI	**Safe stop if 0G is unreachable, no faked decision**	2:40–3:05
0G Private Computer	**Inference in the main user flow, with request IDs**; a reason to come back (recurring jobs, scored robots)	1:45–2:25, 3:35
Arkiv	**A feature users see reading Arkiv data** (ranked robots, orders, What went to Arkiv)	0:40, 3:05
All	Under the time limit: 0G asks for **under 3 min**. Cut this 4-minute version to 2:55 by dropping 2:25–2:40 (extras) and 3:35–3:50 (scores table) and shortening the montage	

Also needed with the submission: project name and description, public repo with setup, live demo link, contract addresses (none deployed; Arkiv contract and wallets listed in the README), team: Semmal, Telegram/X jonathan_vineet.

